සියලු ම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம் இலங்கைப் பரீட்சைத் திணைக்களம்
Department of Examinations, Sri Lanka Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2023(2024)
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2023(2024)
General Certificate of Education (Adv. Level) Examination, 2023(2024)

ව්‍යාපාර සංඛ්‍යානය I வணிகப் புள்ளிவிவரவியல் I **Business Statistics I**	**31 E I**	පැය දෙකයි இரண்டு மணித்தியாலம் ***Two hours***

Instructions:

* *Answer **all** questions.*
* *Write your **Index Number** in the space provided in the answer sheet.*
* *Statistical tables will be provided. Use of calculator is **not allowed**.*
* *Carefully follow the Instructions given on the back of the answer sheet.*
* *In each of the questions **1** to **50**, pick one of the alternatives from (1), (2), (3), (4), (5) which is **correct** or **most appropriate** and **mark your response on the answer sheet with a cross** (×) on the number of the correct option in accordance with the instructions given at the back of the answer sheet.*

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1. Which of the following statements is true about statistics and its scope?

(1) Statistics does not provide a direction for making optimal decisions when an uncertainty exists.
(2) Statistics has an ability to analyse individual experiences scientifically in a broad range.
(3) Statistics does not provide a base for explaining complex scenarios in a simple manner.
(4) The size of any phenomenon can be quantified accurately by statistics.
(5) Giving reliable conclusions from a statistical study completely depends on the primary data.

2. Consider the following statements.

A - Identifying the measurement scales of variables prior to data analysis is not a compulsory prerequisite.
B - The data collected by an individual or an institute for another study can be utilized only if they align with objectives of the considered study.
C - When compared to the personal interview method, the sample size can be increased in the focused group discussion method.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

3. Which of the following statements is **false**?

(1) A pilot survey is used only to test the appropriateness of a questionnaire prepared for data collection.
(2) The most accurate and precise conclusions can be drawn from a sample survey if well-trained enumerators can be employed.
(3) In open questions of a questionnaire, the respondent has freedom to provide answers independently.
(4) The complete enumeration is an effective approach to maintain a higher accuracy if constraints are manageable.
(5) A complete enumeration provides the chance to obtain an updated sampling frame, which is useful for future surveys.

[see page two

4. Consider the following statements about data presentation methods.

A - Presenting data using charts is one way of performing data analysis.
B - Graphical methods of data presentation can be used to identify the different patterns of data.
C - Optimal conclusions can always be drawn from the results obtained from different data presentation methods.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

5. Which of the following data presentation methods can be used to retrieve a data array?

(1) Frequency Polygon (2) Z-chart (3) Lorenz curve
(4) Box-and-whiskers' plot (5) Stem-and-leaf diagram

6. If the mid value of the first-class interval of a non-overlaping frequency distribution for group data is 20, and the difference between two consecutive upper boundaries is five (05), the limits of the second-class are,

(1) 17.5 - 22.5 (2) 18 - 22 (3) 22 - 28
(4) 23 - 27 (5) 22.5 - 27.5

7. Consider the following statements about the measures of central tendency.

A - It provides measures comprised of theoretical properties which can be easily used for statistical inferences.
B - Mean, median and mode are three typical measures that can be employed for arithmetic operations.
C - Mean is less impacted by sample variations compared to the median and the mode.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

8. Consider the following data related to the weight (in kg) of 11 participants in a weightlifting tournament.

90, 55, 60, 91, 84, 93, 70, 66, 60, 58, 78

If one of the participants' actual weight of 63 Kg was recorded incorrectly as 93 Kg, the correction of this error is affected for the

(1) Mean.
(2) Median.
(3) Mean and median.
(4) Mean and mode.
(5) Mean, median and mode.

9. Which of the following statements is **false**?

(1) When the mean is the most appropriate to measure the central tendency of a data distribution, the standard deviation is the most appropriate to measure its dispersion.
(2) Changing the variance of a data distribution does not directly or indirectly impact its skewness.
(3) When the median is the most appropriate to measure the central tendency of a data distribution, quartile deviation is the most appropriate to measure its dispersion.
(4) When the percentage kurtosis coefficient (K) is close to 0.263 for a data distribution, the kurtosis of the distribution is close to kurtosis of the normal distribution.
(5) When the percentage kurtosis coefficient (K) is close to 0.5 for a data distribution, the kurtosis of the distribution is close to kurtosis of a leptokurtic distribution.

[see page three

10. Consider the following stem-and-leaf diagram.

3	4	8								
4	2	5	7							
5	1	2	4	7	9					
6	0	1	2	3	5	6	7	8	9	
7	0	1	2	3	4	5	6	7	8	9
8	0	2	3	4	5	6	7			
9	0	1	3							

If $\frac{1}{2}(P_{90} - P_{10}) = 21$, the percentage kurtosis coefficient (K) for the distribution is,

(1) 0.25 (2) 0.50 (3) 1.00 (4) 2.00 (5) 4.00

11. The median and the quartile deviation for a data distribution are 115 and 6 respectively. If the first quartile (Q_1) for the distribution is 106, Bowley's coefficient of skewness is,

(1) -1.00 (2) -0.50 (3) -0.05 (4) 1.00 (5) 2.00

12. Which of the following statements is true about correlation and regression?

(1) When the correlation coefficient between X and Y is zero it does not mean the non-availability of any relationship between X and Y.

(2) When the regression coefficient of Y on X is greater than 1, the regression coefficient of X on Y also should be greater than 1.

(3) The regression coefficient and the correlation coefficient have opposite signs.

(4) The product of regression coefficients of Y on X and X on Y is equal to the square root of the coefficient of determination.

(5) If the given regression model is $y = \beta_0 + \beta_1 x + u$, only y and x are the random variables.

13. Consider the following statements about the free-hand method of estimating a regression line.

A - It does not have a reliable mathematical base to obtain a regression line.

B - It does not allow for obtaining an infinite number of straight regression lines in a scatter diagram.

C - It requires a scatter diagram for drawing a more appropriate straight regression line.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.

(4) only A and C. (5) all A, B and C.

14. If the sum of the squares of the difference in the corresponding ranks given by two judges P and Q in a debating competition with ten (10) contestants takes the maximum possible value for $\sum d_i^2$, then the rank correlation coefficient between P and Q represents,

(1) Perfect negative relationship. (2) Poor negative relationship. (3) No relationship.

(4) Poor positive relationship. (5) Perfect positive relationship.

15. Consider the following statements about the approaches to probability.

A - The probability that there will be a pandemic situation during the next year in the country cannot be assigned under any probability approach.

B - The probability of an event obtained under the relative frequency approach always be equal to the true probability of the event.

C - The probability of getting a defective item from a large lot of items can only be obtained under the relative frequency approach.

The **false** statement/s from the above is/are,

(1) only A. (2) only B. (3) only C.

(4) only A and B. (5) all A, B and C.

[see page four

16. If $P(X)=0.2$, $P(Y)=0.5$ and $P(X' \cap Y')=0.4$, which of the following statements is **false**?

(1) $P(X \cap Y) > 0$

(2) $P(X \cap Y) = 0.1$

(3) $P(X' \cap Y) = P(X \cup Y)'$

(4) $P(X \mid Y') \neq P(X \mid Y)$

(5) $P(X' \mid Y') = 1 - P(X \mid Y)$

17. If A and B are two events with $P(A)=4P(B)$, $P(A \cup B)=0.8$ and $P(A \cap B)=0.2$, then the value of $P(A \mid B')$ is,

(1) 0.20 (2) 0.25 (3) 0.40 (4) 0.60 (5) 0.75

18. A, B and C work independently on solving a problem. If the respective probabilities that they will solve it are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{2}{5}$, the probability that the problem will be solved by at least one of them is,

(1) $\frac{1}{30}$ (2) $\frac{1}{10}$ (3) $\frac{7}{10}$ (4) $\frac{29}{30}$ (5) $\frac{59}{60}$

19. The following table shows the number of mobile phones sold (X) per month by a person and the probabilities of selling them.

X	10	20	25	30
$P(X)$	0.4	0.3	0.2	0.1

If a certain person receives a basic salary of Rs. 20 000 per month and Rs. 500 for every mobile phone sold per month, then the expected salary he receives each month would be,

(1) Rs. 20 000 (2) Rs. 20 019 (3) Rs. 20 500 (4) Rs. 29 000 (5) Rs. 36 500

20. A multiple-choice question paper has n questions with five choices of answers for each question. If the answer to each question is guessed, then the equation for obtaining the solution of the probability of getting at least one correct answer is greater than $\frac{3}{4}$ would be,

(1) $\left(\frac{4}{5}\right)^n > \frac{3}{4}$ (2) $1-\left(\frac{4}{5}\right)^n < \frac{3}{4}$ (3) $n\left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^{n-1} > \frac{3}{4}$

(4) $1-n\left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^{n-1} > \frac{3}{4}$ (5) $1-\left(\frac{4}{5}\right)^n > \frac{3}{4}$

21. A certain police station receives an average number of 5 calls per hour. The probability of receiving more than 3 calls per 30 minutes is,

(1) 0.1414 (2) 0.2424 (3) 0.4562 (4) 0.5438 (5) 0.7576

22. If the random variable X has a normal distribution with mean 3 and variance 0.25 when $P(3 \leq X \leq c) = 0.4726$, the value of c is,

(1) 0.96 (2) 1.92 (3) 3.00 (4) 3.48 (5) 3.96

23. Consider the following statements about the stratified random sampling.

A - A representative sample cannot be obtained when strata are overlapped.

B - Sampling frame is not required when sampling units are heterogeneous.

C - Managing administrative works of the sampling survey is complicated.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.

(4) only B and C. (5) all A, B and C.

[see page five

24. Which of the following statements is true about the sampling?

(1) When sampling units are in a random order, the results of systematic sampling are equal to the results of the cluster sampling.

(2) In probability sampling, the investigator has an opportunity to select the most suitable sampling units relevant to the objectives of the study.

(3) The probability of selecting a five-digits number from any place in a random number table is 0.000001.

(4) In non-probability sampling, accuracy of conclusions always confirm due to the higher personal bias.

(5) The cluster sampling is suitable for a wide range of studies under the low cost requirement.

25. A simple random sample without replacement was drawn from a population with $N = 246$. If the calculated sample mean $\overline{X}$ and variance S^2 were 210 and 98 respectively, then the standard error of the mean [SE($\overline{X}$)] for a random sample with size $n = 46$ would be,

(1) $\frac{10}{\sqrt{123}}$ (2) $\frac{\sqrt{40}}{7}$ (3) $\frac{70}{\sqrt{23 \times 123}}$ (4) $\sqrt{\frac{40}{23}}$ (5) $\frac{7}{\sqrt{23}}$

26. Which of the following statements is true?

(1) Estimated value and the margin of error are considered in estimating a confidence interval.

(2) The margin of error is the summation of the standard error and the relevant table value.

(3) The standard deviation of the population parameter is called the standard error.

(4) The confidence limits are the end values of a confidence interval for the sample statistic.

(5) Sampling with replacement provides a higher precision for an estimator than sampling without replacement.

27. Consider the following statements about the point estimation.

A - It is a maximum value estimated for an unknown population parameter.

B - It is a single value estimate for an unknown population parameter.

C - It is a range of values estimated for an unknown population parameter.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.

(4) only B and C. (5) all A, B and C.

28. The 97% confidence interval for the mean of a normal population is (116.49, 129.51). If the margin of error is 5.88 at 95% confidence level, the confidence interval is,

(1) (111.48, 134.52) (2) (113.36, 132.64) (3) (117.12, 128.88)

(4) (240.12, 251.88) (5) (234.48, 257.52)

29. If the $Z_{\alpha/2}$ is a part of the equation of the confidence interval for the parameter of a normal population, the $\alpha/2$ is,

(1) the level of confidence.

(2) the level of significance.

(3) the probability of the confidence interval constructed for the population mean.

(4) the summation of areas of lower tail and upper tail of the sampling distribution.

(5) the area in the lower tail or upper tail of the sampling distribution.

30. Consider the following statements.

A - Statistic is a function of sample elements estimated for the unknown parameter.

B - Estimate is a single value obtained by substituting the sample data for an estimator.

C - An estimator with a minimum standard error is called a biased estimator.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.

(4) only B and C. (5) all A, B and C.

[see page six

31. If the 95% confidence interval for the population parameter μ is represented by $(\overline{X} \pm Z_{\alpha/2} SE(\overline{X}))$, the confidence coefficient of the interval is,

(1) 0.025 (2) 0.050 (3) 0.900 (4) 0.950 (5) 0.990

32. A researcher needs to estimate the confidence interval for mean difference of water consumption units of families with 4 members in rural and urban areas. The following information was drawn from two independent samples of two areas related to water consumption with equal population variances.

	Urban	Rural
Number of Families	10	12
Standard deviation	12	10

The pooled variance is,

(1) 108.9 (2) 119.8 (3) 120.0 (4) 124.2 (5) 132.0

33. It is reported that the 50% of students has favoured to the online mode examinations in a previous survey. The required sample size for repeating the survey with a 95% confidence level and a margin of error 0.05 would be approximately,

(1) 192 (2) 196 (3) 384 (4) 392 (5) 768

34. Consider the following statements about the statistical hypotheses testing.

A - A statistical hypothesis is a logical assumption considered to be temporarily true about the unknown population parameter.
B - Confidence intervals can be used to test any statistical hypothesis.
C - In hypotheses testing, it is minimized the type I error by keeping the type II error at a certain constant level.

The true statement/s from the above is/are,

(1) only A. (2) only A and B. (3) only A and C.
(4) only B and C. (5) all A, B and C.

35. A company distributed 20 ml medicine bottles to pharmacies and received complaints that the liquid contents in bottle is less than the labelled amount. To test this claim, the company selected 64 bottles randomly and revealed that the mean was 19.915 ml with a standard deviation of 0.4 ml. If the *p*-value of this test is 0.0446, the conclusion would be,

(1) reject the null hypothesis at $\alpha = 0.01$
(2) do not reject the null hypothesis at $\alpha = 0.10$
(3) reject the null hypothesis at $\alpha = 0.025$
(4) do not reject the null hypothesis at $\alpha = 0.05$
(5) reject the null hypothesis at $\alpha = 0.05$

36. The number of power-cut times per year in five different areas is as follows.

Area	A	B	C	D	E
Number of power-cut times per year	40	60	30	50	20

To test the randomness of occurrences of power-cuts per year in five areas at $\alpha = 0.05$, the test statistic is,

(1) 12.7 (2) 25.0 (3) 32.0 (4) 50.0 (5) 100.0

37. Which of the following statements is true about the test statistic?

(1) A decision about the null hypothesis is taken by comparing the test statistic with the relevant table value.
(2) In testing a hypothesis about the mean of a normal population with known population variance, the test statistic is derived using the *t*-distribution.
(3) When the hypothesis is about the mean of a normal population, the test statistic is calculated based on only the sample size.
(4) A test statistic about the population parameter is calculated using all population observations.
(5) When hypotheses testing is about the difference between means of two populations, the test statistic is always derived assuming variances of two populations are equal.

[see page seven

38. A sociologist wanted to determine the relationship between gender and the preference about smartphone brands. If the given contingency coefficient value and the chi-square test statistic were 0.2 and 6.72 respectively, the total number of observations considered was approximately,

(1) 123 (2) 134 (3) 161 (4) 175 (5) 181

39. A data analyst wanted to investigate the effectiveness of three different teaching methods. He assigned five (5) students each randomly to each teaching method and measured students' examination performance. If the calculated F-test statistic and the mean squares of errors (MSE) were 15.5 and 10 respectively, the sum of the squares of totals (SST) would be,

(1) 350 (2) 430 (3) 450 (4) 640 (5) 805

40. Consider the following statements about the time series analysis.

A - The line graph is the most appropriate method of depicting time series data for identifying different movements.

B - The declining gross domestic production (GDP) growth rate of a country during the Covid-19 pandemic is associated with the characteristic of seasonal movements.

C - When original data of a time series are expressed as a percentage of the moving average, seasonal and irregular movements will be removed.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

41. Which of the following statements is **false?**

(1) Personal bias occurs when determining the order for moving averages of a time series with annual data.
(2) Since cyclical and irregular movements are removed, the moving average method can be used to obtain more precise forecasting values.
(3) The least square trend equation can be used to obtain the most reliable forecast values.
(4) The additive model is appropriate when components of the time series are depend on each other.
(5) When only the long-term trend is not available in a time series, the average percentage method is appropriate for estimating seasonal indices.

42. When the original time series value (Y) for each month is expressed as a ratio of the estimated corresponding moving average value, the seasonal indices can be obtained by,

(1) dividing by the trend (T) component and taking the monthly average over successive years.
(2) dividing by the trend (T) component and taking the annual average over successive months.
(3) subtracting trend (T) component and taking the monthly average over successive years.
(4) taking the average value for each year over successive months.
(5) taking the average value for each month over successive years.

43. The annual trend equation estimated for the fertilizer requirement (in thousand Metric Tonnes) of a certain country is given $\hat{Y}=72+0.64t$ (origin-2020). If the seasonal index for the third quarter of 2023 is 120, the estimated fertilizer requirement for the third quarter of the year 2023 is approximately in Metric Tonnes,

(1) 20 000 (2) 22 200 (3) 24 000 (4) 26 300 (5) 28 900

44. The process average was 3.0062 for an inspection process with a sample size of 11. When the upper control limit for the process variability (UCL_R) was calculated as 0.872 the upper control limit of the $\overline{X}$-chart would be,

(1) 2.0354 (2) 2.8637 (3) 3.1487
(4) 3.4927 (5) 3.9769

45. Consider the following statements about statistical quality control.

A - If the causes for variations of a process cannot be directly identified, its corrections can be done by only the relevant quality controller.

B - It can be confirmed a higher quality level under the lower inspection cost by involving well-trained quality controllers.

C - Variations in the process can be identified earlier when pre-determined standards have been established.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only B and C. (5) all A, B and C.

46. Twenty six samples of size 40 were chosen from a production process. Trial control limits were established for the p-chart and CL_p was 0.05. 12 defective items were recorded in the 20th sample and was lied out of the control limits. After eliminating this sample, the new CL_p is,

(1) 0.038 (2) 0.039 (3) 0.040 (4) 0.050 (5) 0.051

47. For a sampling plan with a lot size of 1500 and a sample size of 150, the acceptance number is 4. The probability of accepting the sampling plan at a 4% defective percentage is,

(1) 0.1512 (2) 0.2851 (3) 0.7149 (4) 0.8488 (5) 0.9999

48. Consider the following statements about index numbers.

A - A recent year, without impacts of unusual events, should be selected as the base year.

B - It does not create a problem in calculating index numbers when consumer goods which were selected for the base year are not available in the given year.

C - The Laspeyres' index can be calculated at a lower cost compared to the Paasche's index.

The true statement/s from the above is/are,

(1) only A. (2) only B. (3) only A and B.
(4) only A and C. (5) all A, B and C.

49. If the price relative of the tea will be increased by 25% in the year 2024 by a particular tea exporting company, the percentage of the quantity relative increased in order to double the sales (value relative) in 2024 would be

(1) 25% (2) 30% (3) 60% (4) 62.5% (5) 75%

50. Calculated Laspeyres' and Paasches' price indices considering 2015 as the base year and 2022 as the given year are 106.4 and 105.7 respectively. Then the Marshall-Edgeworth price index for 2022 should be a value,

(1) equal to 106.4
(2) greater than 106.4
(3) less than 105.7
(4) equal to 105.7
(5) between 105.7 and 106.4

* * *

සියලු ම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved]

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2023(2024)
கல்விப் பொதுத் தராதரப் பத்திர (உயர் தர)ப் பரீட்சை, 2023(2024)
General Certificate of Education (Adv. Level) Examination, 2023(2024)

ව්‍යාපාර සංඛ්‍යානය வணிகப் புள்ளிவிவரவியல் **Business Statistics**	II II II	**31 E II**

පැය තුනයි மூன்று மணித்தியாலம் ***Three hours***	අමතර කියවීම් කාලය - මිනිත්තු 10 යි மேலதிக வாசிப்பு நேரம் - 10 நிமிடங்கள் **Additional Reading Time - 10 minutes**

Use additional reading time to go through the question paper, select the questions you will answer and decide which of them you will prioritise.

Instructions:

* *Answer five questions selecting at least two questions from each part.*
* *Statistical tables and graph papers will be provided. Use of calculators is not allowed.*

Part I

1. (a) Describe the following methods of data collection, giving **two** advantages and **two** disadvantages of each.
   (i) Telephone interview method
   (ii) Electronic method (06 *marks*)

   (b) State **two** characteristics of the ratio scale data and compare them with respect to the ordinal scale data. (02 *marks*)

   (c) Explain how the Z-chart can be effectively applied in the decision-making process of a certain company on its monthly sales data. (03 *marks*)

   (d) A self-employed woman makes bags for selling purposes using recycling papers. The number of bags sold in a period of 100 days is given in the following table.

Number of bags sold	Number of Days (f)
35 – 40	02
40 – 45	03
45 – 50	10
50 – 55	15
55 – 60	27
60 – 65	23
65 – 70	13
70 – 75	05
75 – 80	02

   (i) Draw the ogive (less than) for the above data. (03 *marks*)
   (ii) Find the total sales of 12 days with least bags sales by using the ogive. (01 *mark*)
   (iii) If Rs. 325.00 tax per day is charged when the daily bags sales exceeds 68, calculate the total tax that has to be paid approximately by using the ogive. (02 *marks*)
   (iv) Calculate Kelley's coefficient of skewness by using the ogive. (03 *marks*)

[see page two

2. (a) Describe which measures of central tendency are applicable in each of the following situations with reasons.

(i) Selecting the most popular actor for an awarding ceremony.

(ii) Calculating the Z-score for each student by obtaining the Z-values for the relevant subjects.

(iii) Selecting 50% of athletes for the second round of the 400m running event. (03 *marks*)

(b) Explain the roles of measures of dispersion, skewness and kurtosis for a data distribution. (03 *marks*)

(c) Information on age-specific female and male deaths due to the COVID-19 pandemic in a less impacted province are given in the following table.

Age	No. of male deaths (f_M)	No. of female deaths (f_F)
40 - 44	4	2
45 - 49	6	7
50 - 54	9	15
55 - 59	14	10
60 - 64	18	5
65 - 69	9	1
	60	40

The mean, median, mode, variance and Karl Pearson's coefficient of skewness of the age distribution of male deaths are 57.25, 58.43, 61.04, 51.19 and -0.39 respectively.

(i) Calculate the mean, median, mode, variance, and Karl Pearson's coefficient of skewness for the age distribution of female deaths. (07 *marks*)

(ii) Comment on shapes of the age distributions of female and male deaths. (02 *marks*)

(iii) Calculate the combined mean for the age distributions of female and male deaths and comment on the appropriateness of the combined mean instead of individual means. (02 *marks*)

(d) Comment on which age distribution has a higher variability in deaths by using an appropriate measure. (03 *marks*)

3. (a) (i) What is the purpose of the time series analysis? (02 *marks*)

(ii) The quarterly average rainfall data (in mm) recorded from the third quarter of 2019 to the second quarter of 2022 in rainfall data collection centers of a certain province are given in the following table.

2019		2020				2021				2022	
Q_3	Q_4	Q_1	Q_2	Q_3	Q_4	Q_1	Q_2	Q_3	Q_4	Q_1	Q_2
100	160	130	110	145	170	160	140	175	220	190	170

Draw the time series plot for average rainfall data and describe time series movements depicted. (03 *marks*)

(b) The following table provides the percentage values of the ratio to the moving average of quarterly sales (in Rs. million) of a particular tea exporting company from 2016 to 2019.

Quarter	2016	2017	2018	2019
Q_1	-	126.9	137.2	147.8
Q_2	-	103.7	76.5	86.2
Q_3	67.6	65.4	54.5	-
Q_4	111.2	106.6	116.4	-

(i) Calculate seasonal indices for each quarter and comment on them. (03 *marks*)

[see page three

(ii) If estimated trend values of the sales of four quarters in the year 2020 are 95.5, 100.6, 105.1 and 111.8 respectively, estimate tea sales for the second and fourth quarters of the year 2020 assuming a multiplicative model. (02 *marks*)

(c) (i) What is meant by giving weight with respect to the index numbers? (01 *mark*)

(ii) State **two** advantages and **two** disadvantages of using base year weight and given year weight in index number calculations. (04 *marks*)

(iii) Explain **four** limitations of the index numbers. (02 *marks*)

(d) (i) State **two** properties of the Marshall-Edgeworth price index. (01 *mark*)

(ii) Retail prices and quantities purchased in four selected consumer goods for a week of the years 2017 and 2019 are given in the following table.

Consumer good	Unit	2017		2019	
		Price	Quantity	Price	Quantity
Fresh milk	1 litre	100	10	200	8
Bread	1 loaf	60	12	110	16
Eggs	box with 4 eggs	60	6	120	8
Dhal	1 kg	150	4	320	2

Calculate the Marshall-Edgeworth price index for the year 2019 and comment on it. (02 *marks*)

4. (a) In regression analysis, describe the forecasting accuracy of Y values, within (interpolation) and out of the given range (extrapolation) of X values. (01 *mark*)

(b) The following table provides the weight (in thousand Kg) of eight different types of motor cars (X) and the fuel (in gallons) needed approximately to drive 100 Km (Y).

Motor Car	A	B	C	D	E	F	G	H
X	1.5	1.8	2.2	0.9	1.5	1.2	0.6	1.3
Y	08	09	11	09	10	08	06	09

$$\sum X = 11,\ \sum Y = 70,\ \sum X^2 = 17,\ \sum Y^2 = 628,\ \sum XY = 100$$

(i) Calculate the product moment correlation coefficient between X and Y, and comment on it. (02 *marks*)

(ii) Estimate the regression line of Y on X by the method of least squares and interpret the coefficients. (04 *marks*)

(iii) Calculate the coefficient of determination and comment on goodness of fit. (01 *mark*)

(iv) When the weight of a car is 2700 Kg, how many gallons of fuel does it need to drive 100 Km? Comment on the forecasting accuracy. (02 *marks*)

(c) (i) What is meant by the quality of a product? (01 *mark*)

(ii) A recently established smartwatch manufacturing company needs to supply high-quality products including batteries with long charged time. The company selected 10 samples of four watches randomly and calculated means and ranges of charged batteries (in hours) of each sample. The relevant data are given in the following table.

Sample number	1	2	3	4	5	6	7	8	9	10
$\overline{X}$	32	47	27	38	20	35	24	25	31	31
R	22	8	8	5	10	6	2	8	20	11

Construct the $\overline{X}$-chart and the R-chart for the above data. Examine and comment whether the process is statistically in controlled. (04 *marks*)

(d) (i) State two uses of the operational characteristics (OC) curve in quality control. (01 *mark*)

(ii) Complete the given table for the acceptance sampling plan with $n=50$ and $c=3$, by considering the following two situations.

1 - A batch of 1000 units including 140 defective items

2 - A batch with 2% defectives

Batch	Defective proportions	Is the lot good or bad?	The probability of accepting the lot	Who is possible exposed to risk?	The risk probability
1					
2					

(04 *marks*)

Part II

5. (a) (i) What is meant by collectively exhaustive events? (02 *marks*)

(ii) "All collectively exhaustive events are mutually exclusive". Comment on this statement with a suitable example. (02 *marks*)

(iii) If A and B are two events such that $P(A|B)=1/4$, $P(B|A)=1/3$, $P(A\cup B)=1/4$ and $P(A\cap B)=p$, then find the value of p and hence find $P(A)$. (03 *marks*)

(b) A school committee of three (3) people is to be selected from a group of four (4) students and two (2) teachers. If the random variable Y represents the number of students included in the committee,

(i) assign the values for the random variable Y. (01 *mark*)

(ii) calculate the probabilities of having above each of the assigned values for Y. (03 *marks*)

(c) A box of 15 ice cream cups contains six (6) chocolate ice cream cups and nine (9) vanilla ice cream cups. Amali chooses an ice cream cup randomly and eats it. Later she eats another one in the same manner. Find the probability that Amali eats,

(i) two chocolate ice cream cups (02 *marks*)

(ii) ice cream cups of both types (02 *marks*)

(d) A teacher in a particular school developed a method for predicting how a student will success on an examination. If the student passes the previous examination, there is a 60% chance of passing the next. If the student fails the previous examination, there is a 50% chance of failing the next. If the student has a 70% chance of passing the first examination, then;

(i) draw a tree diagram of the first three examinations that the student is going to sit (01 *mark*)

(ii) what is the probability that the student passes the first examination but fails the third examination? (02 *marks*)

(iii) what is the probability that the student has failed the first examination, given that the student failed the second examination? (02 *marks*)

6. (a) Describe the conditions that are needed to be satisfied to apply binomial distribution with an example. (02 *marks*)

(b) The wholesale vendor in a village packed boxes containing 100 mangoes in each for transportation to the city for retail. The probability of damage to a mango during transport is 4%. The lot with less than 3% damage only is accepted by the retail vendor.

(i) Define the corresponding random variable which is suitable for the above description. (01 *mark*)

(ii) State the probability distribution of the random variable with its parameters. (02 *marks*)

(iii) What is the probability of getting the lot rejected? (03 *marks*)

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(c) (i) Describe **two** usages of poisson distribution. (02 *marks*)

(ii) Suppose X has a poisson distribution with mean λ_1 and the standard deviation λ_2. If the mean of this distribution is given as 4 then find $P(2X-2 \geq \lambda_1+\lambda_2)$. (02 *marks*)

(d) (i) What is the difference between the normal distribution and the standard normal distribution? (02 *marks*)

(ii) The height of a certain type of plant follows a normal distribution with a mean of 64cm and a standard deviation of 6cm. The height of each plant is measured and has categorized into three grades depending on their height. The top 10% receive "*A*" grade, the next 10% receive "*B*" grade and the third 10% receive "*C*" grade. Find the height ranges of *A*, *B* and *C* grade plants. (06 *marks*)

7. (a) Differentiate between the following pair of terms related to the sampling.

(i) Target population and sample population

(ii) Selection bias and measurement bias

(ii) Sampling fraction and finite population correction factor (06 *marks*)

(b) (i) What is meant by systematic sampling method? Explain under what condition the systematic sampling results are equal to the results of simple random sampling. (02 *marks*)

(ii) Describe the purposive sampling method by giving a merit and a demerit. (02 *marks*)

(c) The following table provides two sampling distributions of $\bar{y}_{WR}$ and $\bar{y}_{WOR}$ obtained from two simple random samples of size 2 (n=2) with replacement and without replacement from a population of size four (N=4) with the values of y_i are 3, 9, 5, and 7.

$\bar{y}_{WR}$	3, 6, 4, 5, 9, 7, 8, 6, 5, 6, 4, 7, 7, 5, 8, 6
$\bar{y}_{WOR}$	6, 4, 5, 7, 8, 6

(i) Calculate the population mean $\bar{Y}$ and population variance S^2. (02 *marks*)

(ii) Calculate expected values and variances of $\bar{y}_{WR}$ and $\bar{y}_{WOR}$ (03 *marks*)

[Use the formula $V(\bar{y}) = \frac{S^2}{n}\left(\frac{N-n}{N-1}\right)$].

(iii) State the most efficient estimator for the population mean $\bar{Y}$ with reasons. (02 *marks*)

(d) The manager of a pizza restaurant mentioned that the time taken for their deliveries around the city are distributed with a mean of 45 minutes and a standard deviation of 6 minutes. Therefore, he claims that their delivery process is operating efficiently. If the mean of a sample with nine (9) deliveries in a selected day is 48 minutes, confirm whether the delivery process is efficient at the 95% confidence level by giving the assumptions. (03 *marks*)

8. (a) (i) Explain what is meant by testing a statistical hypothesis with an example. (02 *marks*)

(ii) Describe the importance of the size of the test and the power of the test in statistical hypothesis testing. (02 *marks*)

(iii) Define the critical value and state **three** factors to be considered in determining the critical value. (02 *marks*)

(b) (i) The manager claimed that the average daily cost (in Rs. thousands) of his taxi service is Rs. 265 and the daily cost cannot be covered after the fuel price revision. Examining a random sample of 25 days, he confirmed that the average daily cost and the standard deviation are Rs. 256 and Rs. 25 respectively. Test whether the manager's claim is acceptable at a 5% level of significance, assuming the daily cost follows a normal distribution. (Consider that H_1: $\mu < 265$). (03 *marks*)

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(ii) The manager of the taxi service is not satisfied with the above test results and asks to test his claim again by increasing the sample size. In examining a sample of 36 days, it was revealed that the average daily cost is Rs. 260 and the standard deviation is Rs. 24. Retest at the same level of significance, whether the manager's claim can be accepted. (03 *marks*)

(c) A researcher wants to investigate the effectiveness of three different teaching methods (M1, M2, and M3) on students' performance at an online-based short course. He assigned four students each to follow the course under each teaching method. The recorded performance marks based on the online examination after completing the course are given in the following table.

Performance marks of students		
M1	M2	M3
12	15	10
13	12	11
14	16	10
11	15	11

$$\sum X_{ij}^2 = 1922$$

(i) State hypotheses for testing whether the performances of students are statistically different under three different teaching methods. (02 *marks*)

(ii) Calculate the test statistic to test the null hypothesis. (04 *marks*)

(iii) State the conclusion about the above null hypothesis at the 5% significant level. (02 *marks*)

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